Ch 7.1-7.2: Polynomial regression and Step Functions

Lecture 21 - CMSE 381

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Announcements

Last time:

- Test 2

This lecture:

- 7.1 Polynomial regression
- 7.2 Step functions

Announcements:

- HW #6 due Sunday 11/3
- Project stuff

Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
23	Fri	11/1	Regression Splines	7.4	HW #7 Due Sun 11/3
24	Mon	11/4	Decision Trees	8.1	
25	Wed	11/6	Class Cancelled (Dr Munch out of town)		
26	Fri	11/8	Random Forests	8.2.1, 8.2.2	HW #8 Due Sun 11/10
27	Mon	11/11	Maximal Margin Classifier	9.1	
28	Wed	11/13	SVC	9.2	
29	Fri	11/15	SVM	9.3, 9.4	HW #9 Due Sun 11/17
30	Mon	11/18	Single layer NN	10.1	
31	Wed	11/20	Multi Layer NN	10.2	
32	Fri	11/22	CNN	10.3	HW #10 Due Sun 11/24
33	Mon	11/25	TBD: Unsupervised learning/clustering	12.1, 12.4?	
	Wed	11/27	Virtual: Project office hours		
	Fri	11/29	No class - Thanksgiving		
	Mon	12/2	Review		

Section 1

Polynomial Regression

Polynomial regression

Replace linear model

$$y_i = \beta_0 + \beta_1 x_1 + \varepsilon_i$$

with

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

- Can learn with linear models by passing in predictors x_i^ℓ
- Tend to not go higher than degree 3 or 4 because makes overly flexible

Example: Wage Data

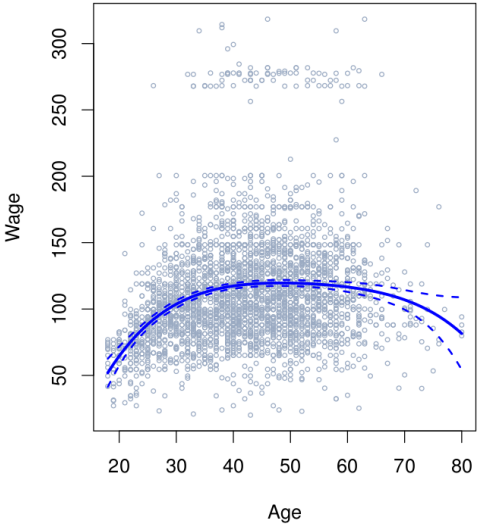
$$\texttt{wage} = \beta_0 + \beta_1 \texttt{age} + \beta_2 \texttt{age}^2 + \cdots + \beta_p \texttt{age}^p + \varepsilon.$$

My code learned:

$$-184.1542 + 21.24552 \cdot age + -0.56386 \cdot age^2 + 0.00681 \cdot age^3 + (-3 \cdot 10^{-5}) \cdot age^4$$

- Make a matrix of predictors X with columns $X^0, X^1, X^2, \cdots, X^p$
- Then can use linear regression to learn the coefficients
- Plot of wage vs age for men in central Atlantic region of the US
- Equivalent figure from the book on the next page

Example with wage data



- Dark line is degree 4 polynomial
- Have variance for each coefficient
- Assume you have a model $\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0 + \cdots + \hat{\beta}_d x_0^d$
- Can use that to estimate the pointwise variance $\mathrm{Var}(\hat{f}(x_0))$
- Draw 2 std deviations away from the line, 95% confidence interval

Section 2

Step function

Step functions

$$\mathrm{I}(X < c) \qquad \mathrm{I}(c_1 \leq X < c_2) \qquad \mathrm{I}(c \leq X)$$

- Draw each of the functions above

More on step function setup

$$
\begin{aligned}
C_0(X) &= I(X < c_1), \\
C_1(X) &= I(c_1 \leq X < c_2), \\
C_2(X) &= I(c_2 \leq X < c_3), \\
&\vdots \\
C_{K-1}(X) &= I(c_{K-1} \leq X < c_K), \\
C_K(X) &= I(c_K \leq X),
\end{aligned}
$$

- Choose values $c_1, \cdots, c_K$
- Allow to learn models that don't have global structure
- Use indicator functions to break up the range of X into bins, then we can fit a new constant in each bin.
- these are sometimes also called dummy variables
- Note that $\sum_j C_j(X) = 1$ because X is in only one interval

Example

Given knots $c_1 = 3$, $c_2 = 5$, $c_3 = 7$, determine the entries in the columns for $C_i(X)$ in the below matrix.

x	$C_0(X)$	$C_1(X)$	$C_2(X)$	$C_3(X)$
1				
2				
3				
4				
5				

x	$C_0(X)$	$C_1(X)$	$C_2(X)$	$C_3(X)$
6				
7				
8				
9				
10				

Draw the function

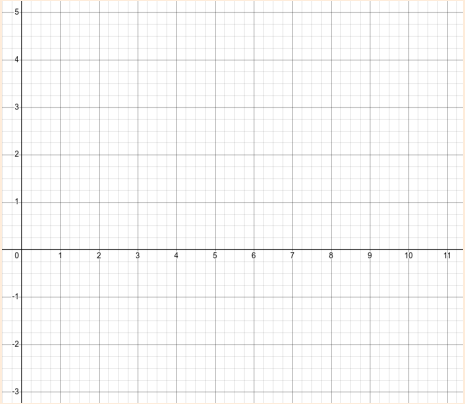
My code doing regression on the step function input returned the function.

$$f(X) = -1 + 3C_1(X) + 4C_2(X) - 2C_3(X).$$

Fill in the table of values, then draw this function below.

X	$F(X)$
1	
2	
3	
4	
5	

X	$F(X)$
6	
7	
8	
9	
10	



Step function: Learned model

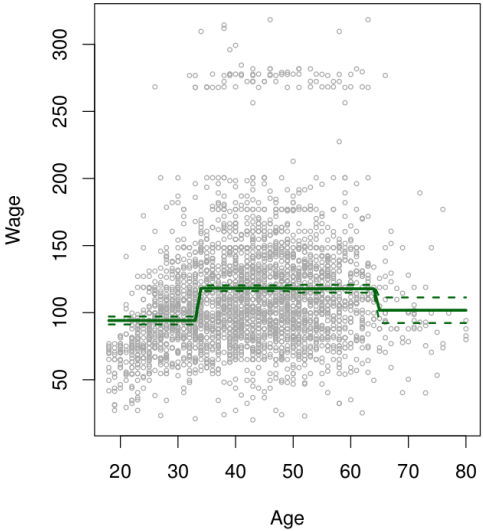
$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$$

- Note above we don't learn a coeff for C_0 since like qualitative variable version, we cna figure out hte value from the others.
- If $X < c_1$, all predictors are 0 so β_0 is mean value of Y for $X < c_1$
- Then response for $X \in [c_j, c_{j+1})$ is $\beta_0 + \beta_j$, so β_j is the avg increase in response for X in the interval relative to $X < c_1$

Coding bit

Back to the `wage` data set

Step function example



- Learned peak where the high earners are

Next time

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	Wed	12/4	**Midterm #3**		